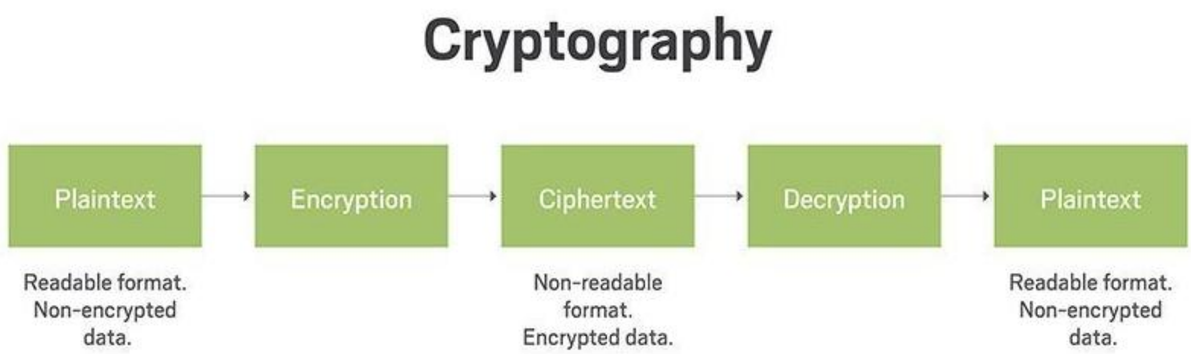
Cryptography Concepts:

- Network traffic is protect, traversing on Internet through cryptographic methods.
- The term cryptography is a Greek word, which means the "Secret Writing".
- “Kryptos,” means hidden or secret and “Graphein” means the Writing.
- Cryptography involves the process of encryption and process decryption.
- Cryptography is the study and practice of hiding data and information.
- Cryptography is the art and science of making data impossible to read.
- Cryptographic algorithms start with plaintext & scramble to unreadable ciphertext.
- Encryption algorithm specify how ciphertext can be decrypted back into plaintext.
- Cryptography is essential part of providing confidentiality, integrity, & authentication.
- Cryptography is a way to keep messages and other data secret from hackers.
- Caesar cipher encryption algorithm is one of the simplest encryption algorithms.
- In this, algorithm every alphabetical character in the plain text is replaced.



Cryptography Terminologies:

Below are some important terminologies to understand Cryptography.

Plaintext:

- The information in its original form or readable data also known as cleartext.
- Plaintext is a text, in natural readable form; it is the data before it is encrypted.
- In simple words, Plaintext or cleartext is the original message or data or info.

Ciphertext:

- An encrypted message is called cipher text, Ciphertext is encrypted text.
- Ciphertext is unreadable until it has been converted into plain text.
- Sometimes it has the same size as plaintext or can be larger than plaintext.
- Cipher text is unreadable by anyone except the intended recipients only.
- Cipher text is the scrambled message or data produced as output.

Encryption:

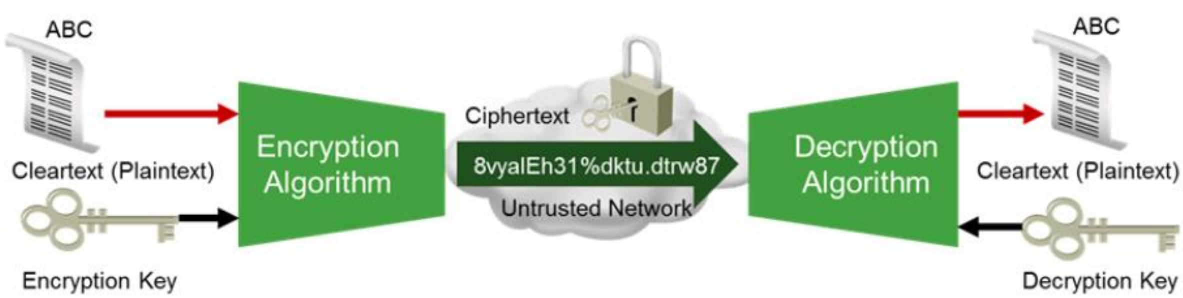
- The process of changing the plaintext into ciphertext is called encryption.
- The process of converting plaintext into the ciphertext is called encryption.
- The encryption is the translation of data into a secret code.
- Encryption is two-way function; encrypted can be decrypted with proper key.
- Encryption is a two-way function that includes encryption and decryption.
- Encryption is the most effective way to achieve data security.
- The main idea of encryption is to protect data from an unauthorized person.
- Encryption needs an algorithm called a cipher and a secret key.

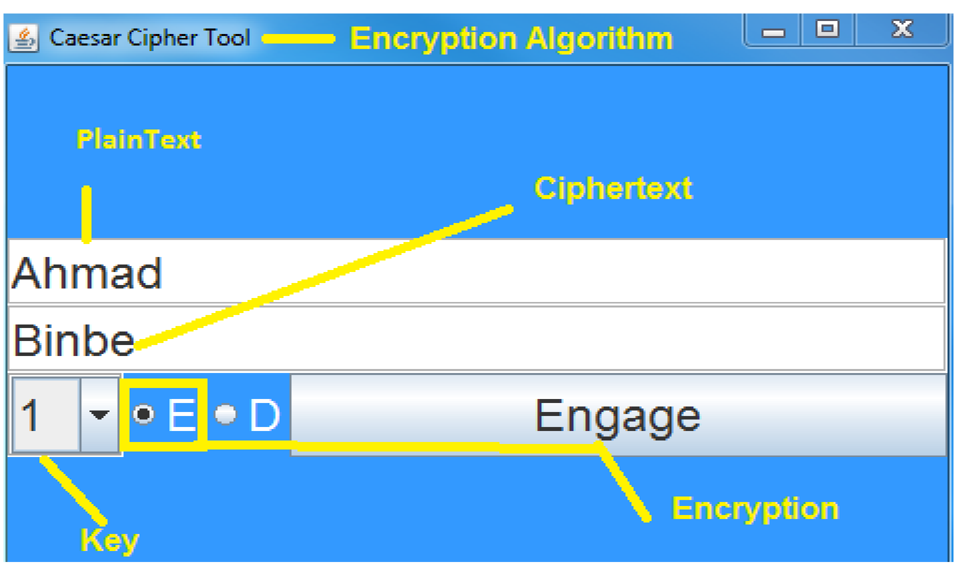
Decryption:

- The process of changing the ciphertext into the plaintext is called decryption.
- The process of converting cipher text back to the original plaintext is decryption.

Encryption Algorithm:

- Algorithm defines how data is transformed when plaintext data scrambled to ciphertext.
- Both data sender & the recipient must know the algorithm used for data transformation.
- Recipient should use same algorithm to decrypt ciphertext back into original plaintext data.





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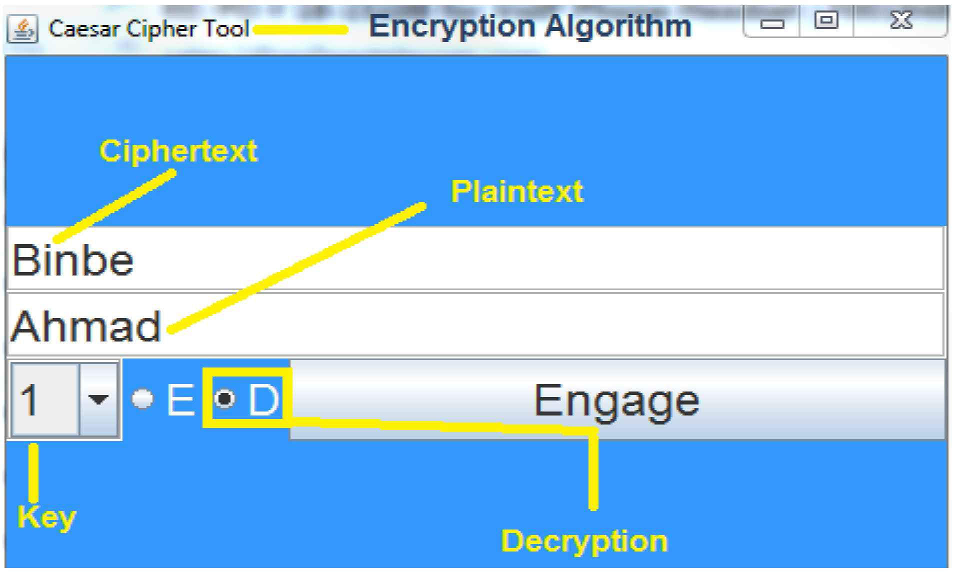


Describe Hash:

- Cryptographic hash function is kind of algorithm that can be run on a piece of data.
- The produced value is called checksum, digest, message digest or hash value.
- It is string value, which is the result of calculation of a Hashing Algorithm.
- Hash is a number that is generated from the text through a hash algorithm.
- Hashing is one-way function that scrambles plaintext to produce unique message digest.
- There is no way to reverse the hashing process to reveal the original data or message.
- Hash Values have different uses main uses is to protects the integrity of your data.
- Protects data against potential alteration so that your data is not changed one bit.
- Sender send the data and digest; receiver takes data & runs its own hash to create digest.
- Receiver then compares digests, if the two digests are same, then data is not manipulated.
- Two methods commonly available MD5 (Message Digest) and SHA (Secure Hash Algorithm).

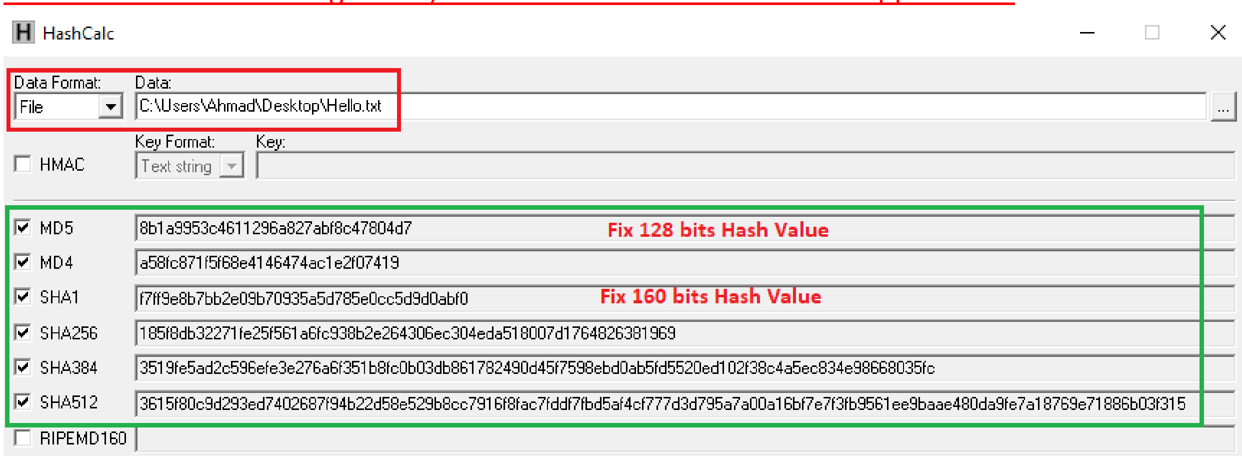
MD5 Hashing:

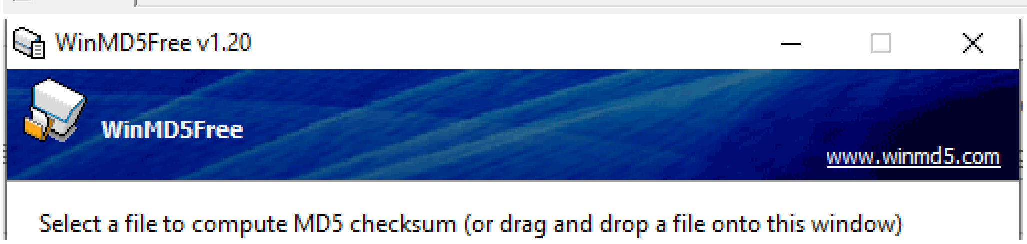
- Hashing is the technique to ensure the integrity.
- MD5, which stands for Message Digest algorithm 5.
- The Message Digest (MD5) is a cryptographic hashing algorithm.
- MD5 hash is typically expressed as a 32-digit hexadecimal number.
- MD5 or message digest algorithm will produce a 128-bit hash value.
- Input data can be of any size or length, but the output size is always fixed.
- MD5 algorithm generates a fixed size (32 Digit Hex) MD5 hash.
- The hash is unique for every file irrespective of its size and type.

SHA Hashing:

- SHA, stands for Secure Hash Algorithm, is cryptographic hashing algorithm.
- SHA used to determine the integrity of a particular piece of data.
- The Secure Hashing Algorithm comes in several flavors.
- SHA-1 and SHA-2 are two different versions of that algorithm.
- SHA1 produces a 160-bit (20-byte) hash value.
- SHA2 has option to vary digest between 224 bits to 512 bits.
- SHA224 produces a 224-bit (28-byte) hash value.
- SHA256 produces a 256-bit (32-byte) hash value.
- SHA384 produces a 384-bit (48-byte) hash value.
- SHA512 produces a 512-bit (64-byte) hash value.

For MD5 and SHA Hashing Demo, use HashCalc and WinMD5 free application.





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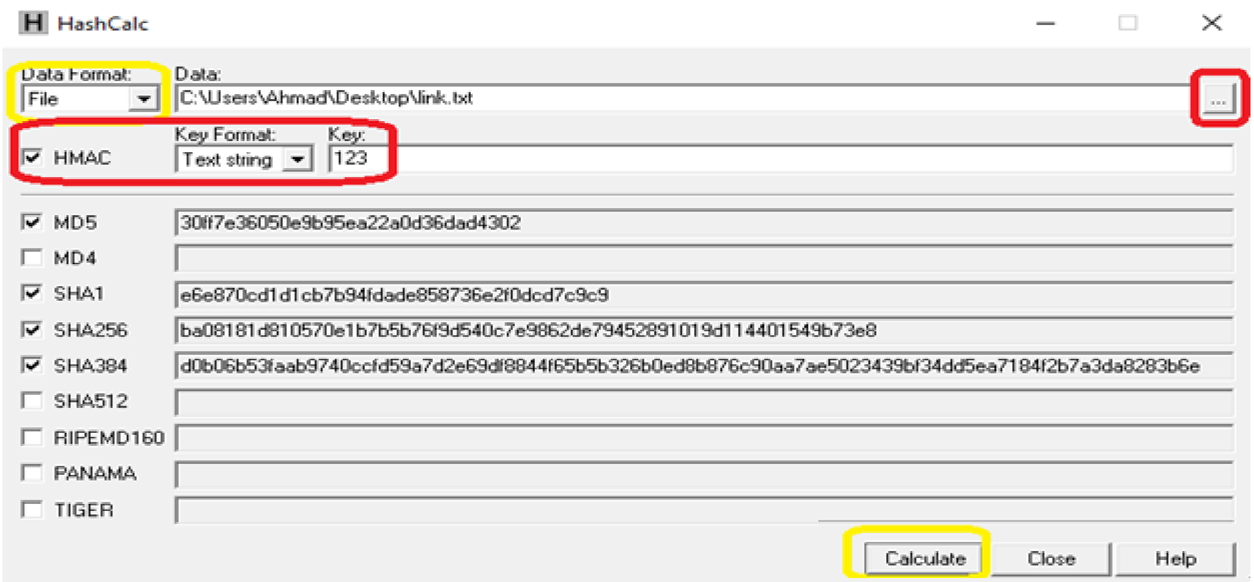
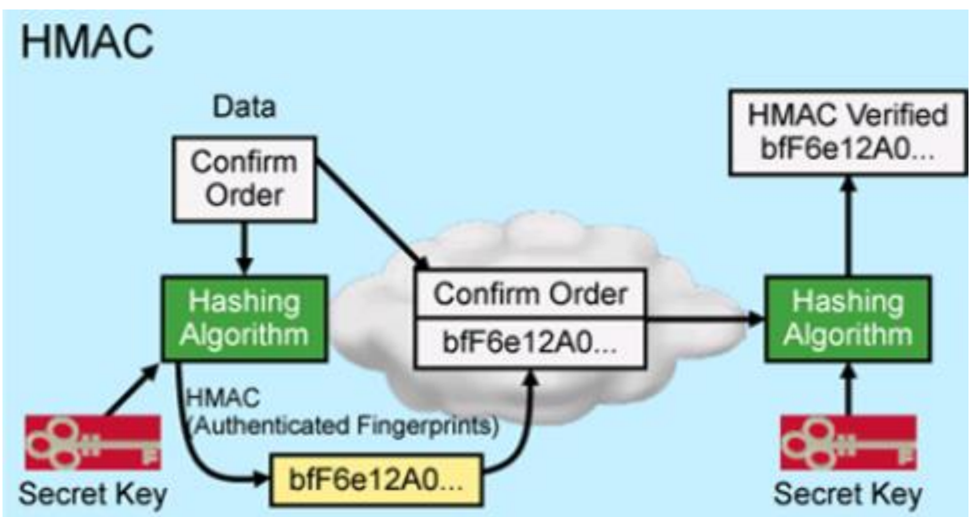
HMAC:

- HMAC stand for Hash Message Authentication Code, Hash with plus secret key.
- HMAC use hash algorithm on data plus secret key that only sender & receiver know.
- Therefore, both parties with secret keys can calculate and verify their hash values.
- This prevents a man in the middle attacks by making it very difficult to modify data.
- This prevents a man in the middle attacks making it difficult and create a new hash.

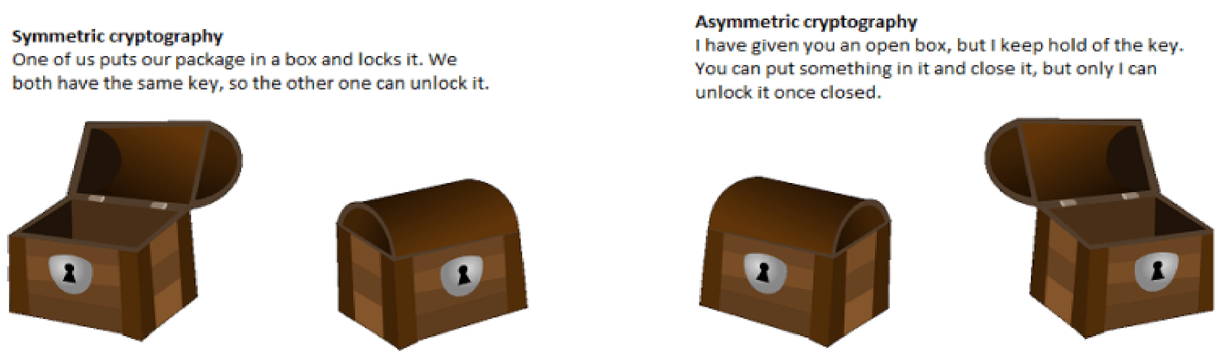




Symmetric and Asymmetric Encryption:

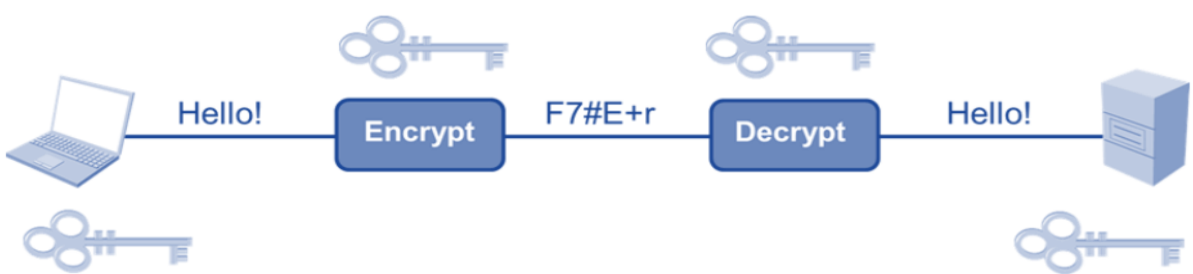
- There are two types of encryption keys symmetric and asymmetric encryption.
- Symmetric encryption uses a single key that needs to be shared among the people.
- Asymmetric encryption uses pair of public key & private key to encrypt and decrypt.
- Symmetric encryption is old technique while asymmetric encryption is relatively new.
- Asymmetric encryption takes relatively more time than the symmetric encryption.
- Symmetric key cryptography utilizes less resource as compared to Asymmetric key.
- Asymmetric key cryptography utilizes more resource as compared to symmetric key.
- An Asymmetric key encryption method is slower than symmetric key cryptography.
- A Symmetric key encryption method is faster than asymmetric key cryptography.

The Simple Lockbox Model



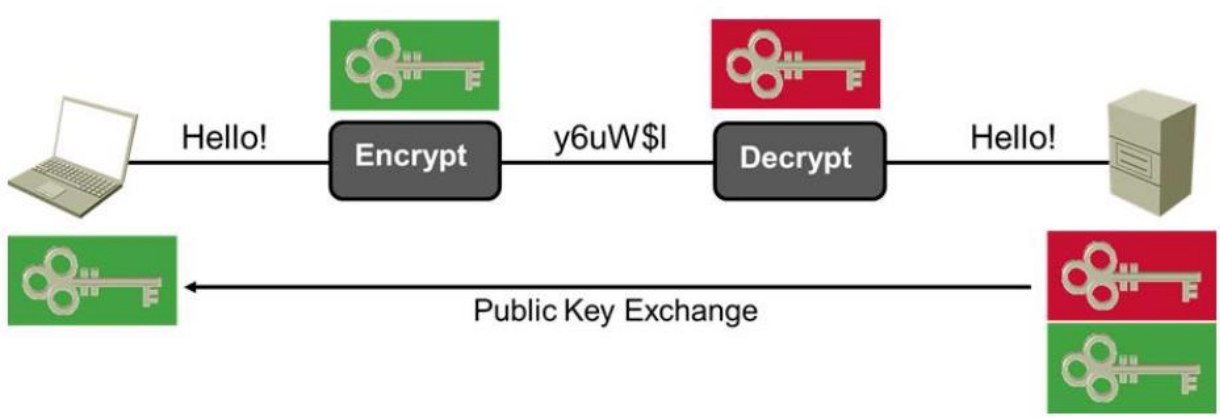
Symmetric Encryption:

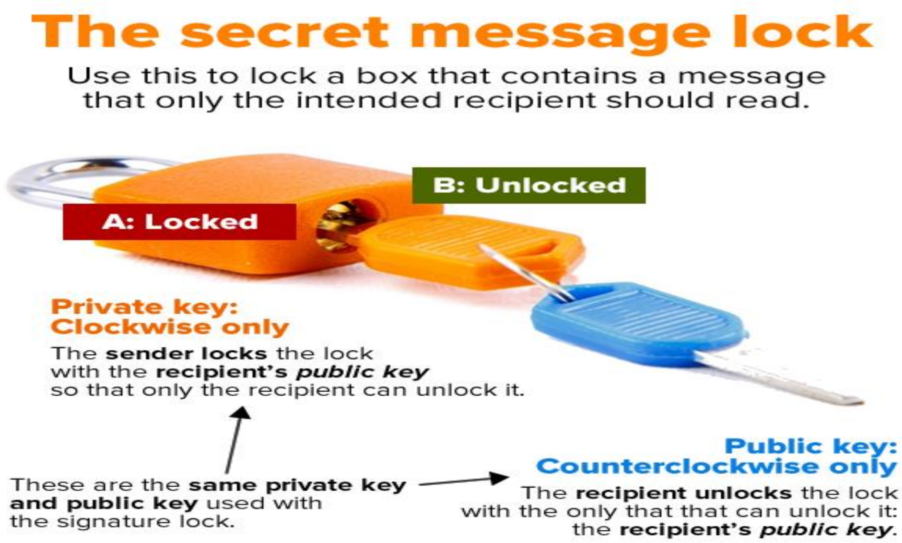
- Symmetric encryption algorithms use the same key for encryption and decryption.
- Symmetric encryption means you use the same key to encrypt and decrypt the data.
- Symmetric key cryptography is called secret key cryptography or private key cryptography.
- Key must be exchanged so that both data sender & recipient can access plaintext data.
- Encryption that involves only one secret key to cipher and decipher information.
- Symmetrical encryption is an old and best-known technique for encryption.
- It uses a secret key that can be either a number, a word or a string of random letters.
- DES, 3DES, AES, IDEA, RC2, RC4, RC5, RC6 & Blowfish are examples of symmetric encryption.
- The most widely used Symmetric Algorithm is AES-128, AES-192, and AES-256.
- The advantage of symmetric encryption is that it is extremely efficient and fast.



Asymmetrical Encryption:

- Asymmetrical encryption is also known as public key cryptography.
- Asymmetric encryption uses two keys to encrypt a plain text.
- A public key is made freely available to anyone who might want to send message.
- The second private key is kept secret so that you can only know.
- A message that is encrypted using a public key can only be decrypted using a private key.
- A message encrypted using a private key can be decrypted using a public key.
- Example of Asymmetrical Algorithms are RSA, DH, DSA, ECC, etc.
- Symmetric encryption is an old technique while asymmetric encryption is relatively new.





<u>For Symmetric and Asymmetric demo, use CrypTool 2.1 Wizard.</u>